NEEDLE: A Live Benchmark of Web Search APIs for AI Agents

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Abstract

AI agents rely on web search APIs, yet no public benchmark tracks how well these APIs retrieve, and a fixed query set goes stale or gets tuned against. We present NEEDLE, an open-source benchmark that builds fresh queries and gold from public sources on every run (one fixed stream excepted), and a measurement study of the 26 search API configurations it has run. Queries come hourly from news feeds and daily from seven public registries and archives. Deterministic matchers or an LLM judge score each result without annotators, and an engine's score is its share of a ceiling pooled from all configurations. On the 12 days when all 21 non-variant engines ran (12.4k query-runs), shares span 15–73% and paired tests separate 178 of 210 engine pairs after Bonferroni adjustment. Over 30 days of scholar runs, 23 of 26 configurations find a paper from its title (recall@10 .71–.98) but spread from .08 to .72 when the query names a detail from its body. Keenable led all eight weeks, yet on one later day of runs, Perplexity edged past it. Blinded LLM raters from three vendors agree with one another at κ .90–.94 and with the judge at .77–.79; no human audited the scorers. The evidence supports choosing a search API by bench and by pairwise tests, not by rank. Code, data and dashboard are public; links are withheld for review.

1 Introduction

An agent that searches for a detail from the body of a recent paper, such as the phrase "gap rent regularity", can read the paper only if its search API returns it. From 29 September to 28 October 2026, Keenable put the paper in its top ten for 72% of such body-keyword queries and ChatGPT search for 8%. Yet both found papers from their titles more than 80% of the time (Figure 1a). Public evidence for choosing between such APIs is thin. IR test collections target document rankers over fixed corpora, and their labels age ("Fed rate decision" changes every six weeks). Agent benchmarks such as BrowseComp score the whole agent, so a weak retriever hides behind a strong model.

NEEDLE measures the search API alone, on five benches (Figure 2). Its unit of evaluation is one search request (retried on errors): a query in, up to k results out, scored on the title, URL and snippet the agent would see. Every run generates fresh queries and gold from public sources, so no fixed test set accumulates (the rare bench and registry facts excepted; Limitations). Where the gold has a canonical identity, such as an arXiv id, a reporter citation or a 10-Q figure, a deterministic matcher checks the results. Where relevance is graded, as for news, an LLM judge rates each result. Each engine's score is a share of a ceiling pooled from all configurations. Scores rest on snippets and, for two benches, on one judge; we audit both with LLMs, not humans (Limitations). We contribute:

- A benchmark that regenerates queries and gold on every run and scores them without annotators. On its two judged benches, blinded LLM raters from three vendors agree with the judge at κ .77–.79 on 1,200 results (§3, Appendix G).
- A normalization by a pooled ceiling that no single engine drives: scoring each engine against a ceiling pooled without it moves its overall share by at most 0.8 points (§3).
- Public results for 26 configurations: over 12 days, overall shares span 15–73% (Table 1), and body-keyword queries separate engines that title queries do not (Figure 1).

The combination is new; each scoring ingredient follows prior work (§4).

2 The Benchmark

Figure 2 shows the five benches. A *configuration* is one search API at one setting: we test 26, from 20 vendors, five of them variants such as a vendor's

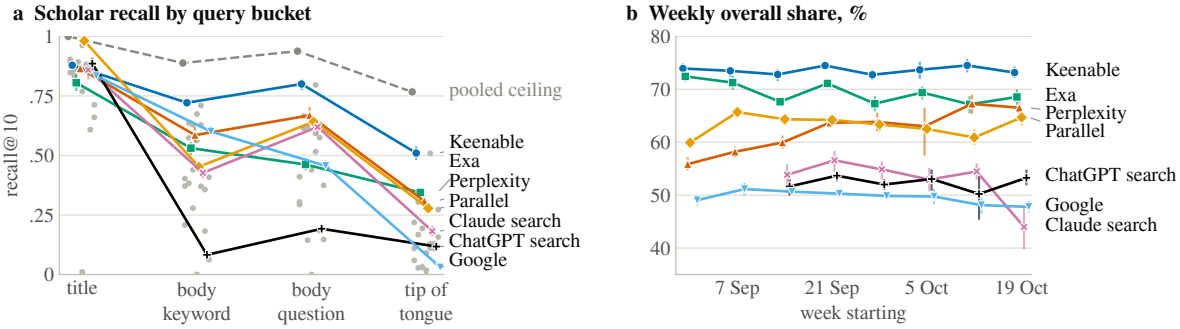


Figure 1: Most engines find a paper from its title but spread apart when the query names a detail from its body (a), and Keenable leads every week while the ranks below it shift (b). (a) Scholar recall@10 per query bucket, 31 runs from 29 September to 28 October 2026, 2,600–3,005 query-runs per line and bucket; lines: the seven engines of (b); grey dots: the other 19 configurations; dashed: the pooled ceiling. (b) Overall share per ISO week, 31 August to 25 October 2026; ChatGPT and Claude search start on 14 September; the pool grew from 18 to 25 configurations. Points: query-weighted means; vertical lines: 95% run-level bootstrap intervals. Scores and failures as the dashboard publishes them (Table 1).

fast or realtime mode. We call a configuration under evaluation an *engine*. A query issued in one run is a *query-run*.

2.1 Query generation

News. Each hour an LLM writes a terse query on the specific event of the newest item from each of 124 feeds and of the Google Trends RSS for 52 US geographies, and refuses evergreen items. Each run samples 20 queries round-robin over topics. **Finance.** Queries ask for registry facts such as a CEO or a ticker (Wikidata, GLEIF), quarterly 10-Q line items (SEC XBRL) or a filing by description. **Scholar.** Each paper yields a *title* query with case and punctuation stripped; a *body keyword* query on a detail absent from title and abstract; a *body question* that asks for the same fact in prose; and a *tip-of-tongue* description with every name and exact value banned. A paper enters only if all four queries pass, so the buckets are paired. **Legal.** Recent opinions from 14 federal courts yield one caption-style query each; the gold is the citation, docket number and CourtListener id. Sections of the Code of Federal Regulations (CFR) yield an LLM query built on one verbatim span of two to eight words, quoted in one variant in four; a query that leaks the citation is rejected (Appendix D). **Rare.** English agent queries from a public stream that contain a word WordPiece splits into five or more pieces. **Operators.** Agent queries often carry operators (Meng et al., 2026; Pezzuti et al., 2026), so at least half of the legal queries and of the finance money and filing queries carry a quoted phrase, `site:`, `after:` or `before:`. The search layer maps each operator to the engine's native filter where one exists (Appendix C).

2.2 Scoring

Graded relevance with a pooled ceiling. For news and rare queries an LLM judge (GPT-6 at temperature 0) sees today's date, the query, a precomputed per-query profile (Appendix B) and one result (title, URL, date, snippet). The judge outputs a 0–4 rating adapted from the Needs Met scale of Google's rater guidelines (Google, 2025), with gates for entity, coverage, freshness and literal operator compliance. Gains are $\{4:1, 3:0.667, 2:0.117, 1:0, 0:0\}$, top-heavy after ERR (Chapelle et al., 2009), and a repeated URL scores zero (Clarke et al., 2008) except on `site:` queries. The judge rates each distinct result once, and every snippet is capped at 2,000 characters. The *ceiling* of a query keeps the best rating per deduplicated URL over every configuration that ran, variants included, sorted by rating; an engine's nDCG@5 takes the ceiling's list as the ideal. A judged query-run is dropped when any of its judgements failed or no configuration reaches a rating of 2. Scholar and legal drop no query for judging, and finance drops an engine's query only when its backstop judgement failed and no result matched (Appendices E and G).

Identifier and answer matching. Scholar and legal scan result URLs and snippets for arXiv ids, DOIs, PMIDs, reporter citations, docket numbers and CFR citations. Recall@k is the fraction of

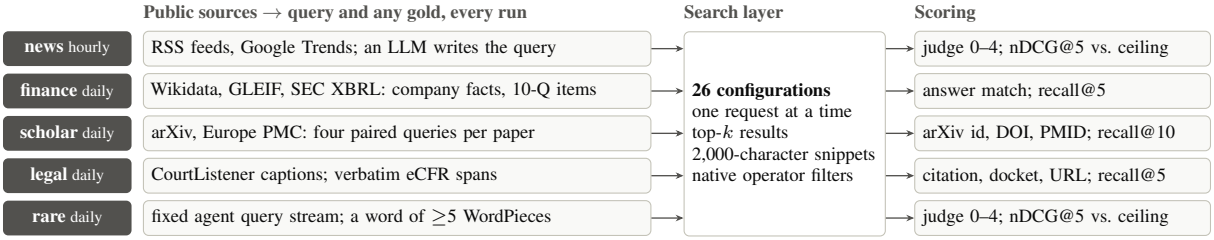


Figure 2: Every run draws its queries, and any gold, from public sources. One search layer sends each query, without its gold, to every configuration; the scorer checks the returned title, URL and snippet against the gold, or the judge rates them.

queries whose top-k results carry the gold identity. For caselaw the identity is the case (opinion, docket page or a citing page), so the metric is case-identity recall (Appendix G). Finance asks whether the top-k snippets *contain the answer*, within 0.1% for money (table figures and parenthesized losses included) and 15% for headcounts (Appendix F). An LLM backstop may upgrade a rejected non-money result, never revoke a hit. Here the ceiling is a hit by any configuration.

2.3 Aggregation

An engine's *overall share* $S(e)$ is the mean, over the five benches b, of its *bench share* $M_b(e)/M_b(\text{ceiling})$, where M_b is the bench metric. $S(e)$ is a share of the best any configuration found, not of the web. Table 1 uses the 276 runs in which all 21 listed engines took part (one per vendor, two for Brave). All scores are the public dashboard's (Appendix A). A failure is classified by its error record: an account-side one (our own quota or rate limit, login or credit) is excluded, and any other (server error, timeout) scores zero, as for an agent. On news and rare, Table 1 scores every failure zero (Appendix A). The exclusion removes up to 28% of an engine's query-runs (TinyFish; Appendix G). Intervals are 95% bootstraps over runs.

3 Results

The judge agrees with LLM raters, and the ceiling is stable. Blinded LLM raters from three vendors (two Claude Opus 5.5 sessions, Gemini 4, Grok 4.7) graded 1,200 judged results from the snippet and the first 3,000 characters of the page. They agree with one another at quadratic-weighted κ .90–.94 and with the judge at .77–.79 (Appendix G). On every result of 100 queries, graded from snippets only, Gemini and Grok rank the 21 engines at Spearman .94 with the judge (95% intervals .89–.97 and .87–.96). These checks show agreement between models, not correctness. Scoring each engine against a ceiling pooled without it moves its overall share by at most 0.8 points.

Shares span 15–73%, and most engine pairs differ. Ceramic is lowest and Keenable highest (Table 1). A run-resampled paired test separates 188 of the 210 engine pairs at 95% and 178 after a Bonferroni adjustment; 164 stay separated under day-level resampling and matched support (Appendix G). After adjustment only Keenable and Ceramic are separated from every other engine; the unseparated pairs form chains of neighbors, such as Exa to Brave (LLM ctx). The lead depends on how the benches are weighted: scholar carries 3.7 of Keenable's 4.7-point lead over Exa, and without scholar the lead is 1.2 points (Appendix G).

Body-keyword queries separate engines that title queries do not. From 29 September to 28 October, 23 of the 26 configurations find a paper from its title (recall@10 .71–.98; Figure 1a, Table 10). On body-keyword queries the same 23 spread from .08 (ChatGPT search) to .72 (Keenable), followed by Keenable's realtime variant (.71) and Kagi (.67).

The leader holds week to week; the ranks below it shift. Over eight weeks, adjacent weekly rankings agree at Kendall's τ_b .81–.94 (1 means the same order), and Keenable leads every week (Figure 1b). Below it, Perplexity passes Parallel from 28 September and Claude search falls by about 13 points, half of it our credit lapse; the growing pool does not drive these moves (Appendix G). On one later day of runs (30 October), Perplexity leads Keenable 72.4 to 71.3 through one scholar and one legal run, each within its bench's run-to-run spread (Appendix G).

A `site:` restriction lowers caselaw recall. In one paired run on 30 October, outside Table 1, adding `site:courtlistener.com` to the same 80 caselaw captions lowers mean recall@5 by .36, from .85. Fifteen of 18 engines

Engine	Overall share, %		Scholar	News	Finance	Rare	Legal	Failed, %	Latency
	mean	95% CI	R@10	nDCG@5	R@5	nDCG@5	R@5	requests	p_{50}, ms
ceiling (pooled)	*100*		*.901*	*1*	*.993*	*1*	*.971*		
Keenable	73.2	72.3–74.1	**.731**	.609	.903	**.508**	**.797**	0.0	135
Exa	68.5	67.4–69.4	.563	.599	**.922**	.476	.771	0.3	2,132
Perplexity	66.9	65.3–68.5	.578	.640	.895	.468	.674	0.0	433
Parallel	64.9	63.6–66.1	.587	.535	.907	.387	.734	0.8	2,660
Brave (LLM ctx)	64.6	63.4–65.6	.533	**.642**	.858	.454	.658	0.0	1,112
Linkup	57.6	56.1–58.9	.470	.567	.797	.400	.570	0.0	1,212
Brave	57.3	56.2–58.4	.488	.557	.769	.354	.621	0.0	487
Context	57.1	55.4–58.6	.598	.469	.782	.339	.576	0.0	866
DeepSeek	56.9	55.8–57.9	.271	.635	.818	.387	.676	0.0	551
You.com	55.8	54.7–56.9	.471	.553	.747	.343	.602	0.0	486
ChatGPT search†§	51.5	49.4–53.1	.300	.340	.809	.366	.697	11.1	43,831
Kagi§	51.3	49.9–52.6	.560	.301	.728	.275	.617	7.6	1,275
Firecrawl	50.7	49.2–52.0	.488	.286	.820	.265	.598	0.0	688
Octen	50.2	48.8–51.7	.307	.511	.730	.322	.586	0.0	197
Google (Serper)	47.7	46.8–48.6	.475	.262	.738	.255	.582	0.0	1,785
Claude search†§	45.1	42.7–47.6	.498	.276	.685	.195	.525	35.9	33,906
TinyFish§	44.5	43.0–45.9	.429	.256	.730	.220	.521	29.0	1,904
Bing (SearchAPI)	44.1	41.1–46.8	.416	.288	.675	.173	.586	0.4	2,008
Jina	43.8	42.4–45.1	.492	.210	.726	.163	.523	4.9	17,272
Tavily	42.6	41.5–43.7	.303	.303	.828	.284	.361	0.1	1,999
Ceramic	14.7	13.9–15.7	.001	.040	.441	.109	.137	0.1	99

Table 1: Overall shares of the ceiling span 15–73%; paired tests separate 188 of the 210 engine pairs, 178 after a Bonferroni adjustment and 164 under two further checks (§3). Data: the dashboard's 276 runs, 17–28 October 2026, in which all 21 listed engines ran (12.4k query-runs; Table 3); the ceiling also pools the variants that ran beside them. †A model's server-side search tool; latency includes answer generation. §Rank depends on excluded account-side failures: scoring them zero or at the ceiling puts Claude search at 34.7–52.0, TinyFish at 37.6–52.0, Kagi at 47.5–53.2 and ChatGPT search at 48.2–54.5 (Appendix G). Bench cells: absolute metric, best in bold; italic: the ceiling. Failed: all failed requests, account-side ones included. Intervals: 95% run-level bootstraps (sampling uncertainty, not scorer error). Latency: per-run median on news.

lose beyond their 95% intervals (Appendix H). Restricting filings to `sec.gov` raises their recall by .35, and CFR sections to `law.cornell.edu` by .25.

These results support choosing a search API by bench and by pairwise tests, not by rank.

4 Related Work

Static test sets leak; live benchmarks regenerate. Static benchmarks leak into training data and get tuned against (Sainz et al., 2023; Balloccu et al., 2024). LiveBench, LiveCodeBench, RealTime QA, FreshQA and LiveSearchBench draw new items from dated sources or Wikidata deltas (White et al., 2025; Jain et al., 2025; Kasai et al., 2023; Vu et al., 2024; Zhou et al., 2025); they score an answering model, and FreshStack scores retrievers over a fixed corpus (Thakur et al., 2025). NEEDLE borrows the idea of fresh items but evaluates live search APIs.

Agents need the retriever measured alone. BrowseComp, GAIA and AssistantBench measure an agent end to end (Wei et al., 2025; Mialon et al., 2024; Yoran et al., 2024). BrowseComp-Plus fixes a corpus to separate retriever from agent (Chen et al., 2026). The closest API study swaps three providers under one agent on a static set (Selvam and Ghosh, 2026), and web search audits and overlap studies compare engines at a few points in time on fixed queries (Lewandowski, 2015; Bevendorff et al., 2024; Kirsten et al., 2026; Yagci et al., 2022). NEEDLE instead scores 26 configurations alone, on queries regenerated every run.

LLM judges work at the system level, with known failure modes. LLM assessors reproduce system rankings at TREC scale (Thomas et al., 2024; Upadhyay et al., 2024b,a; Arabzadeh and Clarke, 2025), so NEEDLE reports engine-level means only; the Limitations discuss the known risks (Clarke and Dietz, 2024; Balog et al., 2025; Alaofi et al., 2024; Soboroff, 2025; Faggioli et al., 2023).

Known-item gold is free, and synthetic queries can rank systems. Simulated known-item queries date to Azzopardi et al. (2007), and LLM-elicited tip-of-tongue queries (Arguello et al., 2021) rank systems like real ones (He et al., 2025). NEEDLE generates queries from documents, as Doc2query, InPars and Promptagator do (Nogueira et al., 2019; Bonifacio et al., 2022; Dai et al., 2023), but adds leak filters. Citation graphs give annotation-free gold for caselaw (Hou et al., 2025),

and full text beats abstracts for retrieval (Lin, 2009); the scholar buckets measure that gap live.

Limitations

Conflict of interest. The authors' organization builds one of the evaluated engines; the final version will name it. Everything that produces the numbers is public: the generator code, the judge prompts, the engine clients, every per-query result with the judge's reasoning, and the workflow that runs the benches. Every engine receives the same queries, the same snippet cap, the same request pacing and the same judge, and each engine keeps one configuration (Table 4) across the Table 1 window (17–28 October). Every benchmark score in this paper is the dashboard's or comes from its scoring code. The benchmark cannot tell whether any vendor, the authors' organization included, changed its index for the benchmark or as ordinary development. Readers should treat the ranking as a claim to audit; the artifacts exist so that they can.

Snippet-only evidence. The judge and the matchers see what the API returned, not the page. That view is what an agent sees before it fetches, and it keeps engines comparable, but it can understate engines whose snippets are terse, and known-item recall misses a target whose URL and snippet lack the gold. Snippet labels can diverge from full-text labels (Kazai et al., 2022), though judging what the reader sees has its own case (Turpin et al., 2009), and LLM judgments from short text track full-text judgments well (Mohtadi et al., 2026). The ceiling inherits the same limit. Engines return snippets of their own length up to the cap, and we treat that length as part of what the news bench measures, since an agent reads the snippet it receives. Appendix G reports the order under an equal snippet budget and checks that known-item hits are the target page.

The LLM judge. News and rare scores depend on one judge model and one prompt. The checks of §3 use LLM raters, not humans. The raters who saw page text read only its first 3,000 characters, with the snippet in the same prompt, and every rater shares the literal-quote rule with the judge, so the checks bound judge noise but cannot rule out blind spots the models share. Query terms in a snippet can fool LLM judges (Alaofi et al., 2024). LLM judges also favor engines built on a similar LLM (Clarke and Dietz, 2024; Balog et al., 2025; Panickssery et al., 2024) and cap measured quality at their own competence (Soboroff, 2025). The judge shares a vendor with ChatGPT search, and the second judge of Appendix G shares one with Claude search. Scholar and legal scores use no LLM, and finance uses one only as a backstop on non-money fields (§2.2), so a judge error reaches the overall share mainly through news and rare.

Finance matcher. Finance recall depends on how the matcher reads money values. Its reading of unit-less table figures (§2.2) cannot tell which period an unlabeled column belongs to, so it can credit a figure for the wrong quarter. Except for the fresh-window audit of that reading, the audits pool engines and so cannot show whether error rates differ by engine. False hits and missed answers need not cancel; Appendix F audits each rule and bounds the effect on every engine.

Coverage and weighting. Queries are English and US-centric (US geographies for trends, US courts and regulations, SEC filers); other languages are unmeasured. The equal weight of the five benches in the overall share is a convention, not an estimate of agent traffic, and Keenable's lead depends on that weighting and does not hold on the one later day of runs, 30 October (§3). Table 1 gives the bench columns for any other weighting, and Table 9 collects the main alternative views of the order. The news bench samples 20 queries per hour, so single runs are noisy; read them through windowed means. Engines run with default settings and a paid key, one configuration per vendor (two for Brave) in Table 1; a vendor's best tier may differ. ChatGPT and Claude search are server-side search tools that retrieve and then generate; we score the sources they return, so they are comparable on quality but not on latency or cost.

Reproducibility. We archive every query, gold value, result and judgment with the judge model and engine list of its run, so anyone can recompute every table from the stored ratings. The archive holds neither the published dates the APIs returned nor each run's code revision, so a rerun of the judge only approximates the original judgments (Appendix G). The next release will stamp each run with its commit.

Gaming and repetition. Query texts can recur across runs on any bench: registry facts by design, and rare queries because we sample them from a

fixed artifact. These repeats are learnable in principle; Appendix G counts them per bench. The news feeds are public, so a vendor could crawl them ahead of the hourly run. A vendor that indexes the sources faster is simply fresher, which is what the news bench measures. The benchmark would not catch a vendor that detected benchmark traffic and answered it differently, and snippets stuffed with query terms would raise judged scores (Alaofi et al., 2024). The per-query artifacts let readers check for either.

Dispatch, vendor changes and failures. The comparison is live but not synchronized: each engine works through the query list in turn, so slow engines see the web later than fast ones. Appendix A measures the lag and its effect; a synchronized dispatcher is future work. Vendor response formats change without notice. In runs after the window, Google (through Serper) and Jina return redirect links instead of result URLs, which the URL-based matchers score as misses (Appendix A). Excluding account-side failures assumes that an engine would have scored on the failed requests as on the rest; without that assumption the ranks of ChatGPT search, Claude search, Kagi and TinyFish are not determined, and Appendix G reports their bounds and the order under matched support. On news and rare, Table 1 scores even our own credit lapses as zero, which lowers Claude search's share (Appendices A and G). A reader of the dashboard sees ranks first; the dashboard will carry these notes beside them.

Ceiling as normalizer. The ceiling is the best any evaluated configuration found, not the best that exists. Adding a strong engine raises the ceiling and lowers every share, so shares are best compared within one window. Unlike a TREC pool, every returned result is judged, so no engine loses credit for unjudged documents (Buckley et al., 2007); the pool only sets the normalizer, and §3 bounds how far an engine's own results raise the ceiling it is scored against (Webber and Park, 2009).

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A Protocol and release

Protocol and scoring rules. This paragraph specifies the dashboard's scorer, which every table in the paper uses unless its caption says otherwise. Results built from run summaries (Table 1, Table 10, Figure 1, the paired tests, bounds and weekly and later-day checks) read the dashboard history as published (`data/history.jsonl` on the `gh-pages` branch). Per-query analyses read the archived artifacts with the scripts in `paper/analysis/`, which import the scoring code of the repository and apply the rules below to every bench. Each failed request is classified from its own error record by `account_side()`: an HTTP 401, 402, 403, 429, 432 or 433 response, or an HTTP 400 response or API error whose message names a credit balance, too few credits, a usage limit, a quota, billing or an API key, is account-side and excluded; every other failure (such as timeouts and server errors) scores zero. A result that carries the gold identity is a hit on any host, `site:` queries included. The finance matcher uses a 0.1% money band with the *table reading* (a scale word after a figure fixes its scale; a bare figure, parenthesized as a loss or not, is read in thousands or millions only when $ or a money

word (*USD*, *dollar*, *millions*, *thousands*, *revenue*, *sales*, *income*, *loss*, *earnings*, *profit*, *EPS*) occurs in the 200 characters before it; never a year, a percentage or a multiple) and without a money backstop; diluted EPS, the only field with cue words, matches only in a result that carries one of them (*EPS*, *per share*, *earnings per*). A repeated URL scores zero except on `site:` queries. The rules live in `src/needle/finance/canon.py` and `src/needle/shared/search/errors.py`. On the dashboard, news and rare runs before 11:05 UTC on 28 October 2026, those of Table 1 included, score every failed request zero, and those from then until 23:47 UTC on 31 October leave every failed request out of the mean; 11 of the 12 scholar runs of Table 1 match DOIs with the scholar matcher that preceded the current one; all other runs apply the rules above, so overall shares computed from the per-query artifacts exceed Table 1 by up to 6.3 points (Claude search). The judge and the query generators are GPT-6 Luna (`openai/gpt-6-luna`) through OpenRouter at temperature 0 with reasoning effort *high*; the second judge is Claude Opus 5.5. Snippets are capped at 2,000 characters before judging and matching. Scholar papers are balanced over five domains (computer, physical, life, social and health sciences) and three ages (7 days, 30 days, 1 year). Table 2 gives each bench's cadence, query count, k and metric.

Bench	Cadence	Queries/run	k	Metric
news	hourly	20	5	nDCG@5 vs. ceiling
finance	daily	120	5	recall@5
scholar	daily	420	10	recall@10
legal	daily	40 + 0–20	5	recall@5
rare	daily	100	5	nDCG@5 vs. ceiling

Table 2: Every engine receives the same k it is scored on. Per-bench protocol. Snippets are capped at 2,000 characters; scholar draws 7 papers per domain and age cell, 4 queries each; legal adds one caption query per opinion the 14 courts published that day.

Rebuilding the tables. Table 3 and `paper/analysis/manifest.json` list, for every headline result, its data, runs, engine set and scoring rules. From the dashboard history (`data/history.jsonl` on the `gh-pages` branch, keeping its rows up to 11:53 UTC on 29 October 2026), two commands rebuild Table 1 with its intervals and the paired-comparison counts of §3:

```
python analysis/table_rows.py \
  history.jsonl \
  table_results.tex \
  analysis/audit/matched_pairs.json
python analysis/adjacent_pairs.py \
  history.jsonl \
  analysis/audit/adjacent_pairs.json
```

Table 1, Table 8 and Figure 1 come from that file as published. The per-query analyses read the archived artifacts of the results dataset and import the scoring code; on the finance, scholar and legal runs of Table 1 they reproduce every dashboard row. Repository commit and dataset revision hashes are withheld for review. Figure 1a comes from the 31 scholar runs of 29 September to 28 October 2026. Every audit in Appendices F and G names its runs and ships its sample and labels under `paper/analysis/audit/`.

API calls. We call each engine through its client in Table 4 at its default tier, one request at a time, with a 30-second timeout and up to three attempts with exponential back-off (1 to 30 seconds) on 429, 5xx and timeouts. We record each request that still fails, with its error type and message. One day of the benchmark issues about 30,700 search requests over the 26 configurations and up to about 75,000 result judgments; a rerun needs the public code, one API key per engine and an OpenRouter key for the judge.

Dispatch order. Within a run the engines are queried concurrently, and each engine works through the query list one request at a time. Across the 249 news runs of 17–29 October that record dispatch times (4,959 query-runs; `dispatch_gap.py`), the first and the last engine to receive the same query are a median of 6.5 minutes apart (90th percentile 12.2 minutes), and three engines create this spread: without Jina, Claude search and ChatGPT search the median is 31 seconds. Over answered requests, the mean lag behind the first engine is under 30 seconds for the other 18 engines, 2.3 minutes for Jina, 4.8 for Claude search and 6.6 for ChatGPT search, so on hourly news the slowest engines see the web later than the fastest. We recomputed news nDCG on the 4,868 scored query-runs of these runs (98,933 engine–query pairs; `lag_subset.py`), keeping only the pairs dispatched within 5 minutes of the first engine. The filter keeps 96% of pairs. The order of the 21 engines is unchanged (Kendall's τ_b 1.0, where 1 means the same order), and no mean nDCG moves by more than .010 (Claude search

Result	Runs	Scored units	Engines
Table 1, Table 8	276 (12 scholar, 232 news, 10 finance, 12 rare, 10 legal), 17–28 Oct	12.4k query-runs	21
Figure 1b	every run in ISO weeks 36–43 (pool grows from 18 to 25 configurations)	weekly query-weighted means	7 shown
Figure 1a	31 scholar, 29 Sep–28 Oct	332–420 queries per run	7 lines, 19 dots and the ceiling
Paired differences (§3)	Table 1 runs; 2,000 run- or day-level resamples	12.4k query-runs	21
Matched support	Table 1 runs (230 of 232 news)	4,048 of these runs' 12,313 query-runs	21
Dispatch lag	249 news runs with timestamps	4,959 query-runs (4,868 scored), 98,933 scored pairs	21
Blinded raters*	17–29 Oct runs with $\geq$19 engines	1,200 results, 200 queries	21, hidden
Second judge*	8 news and 8 rare runs	803 and 566 results	23
List-level raters*	100 complete pools (50 news, 50 rare) from Table 1 runs	5,698 results	21, hidden
Rare quoted vs. unquoted; bench weights; scholar strata	Table 1 runs	1,200 rare query-runs; 5 benches; 4,820 scholar query-runs	21
Caselaw identity*	10 legal runs, 18–28 Oct	100 hits	25
Finance audits*	24 finance runs, 29 Sep–28 Oct	399 results (140 of them exact-band hits), 100 misses	21

Table 3: The data behind each result: the runs it uses, what it counts, and which engines it covers. Every result uses the dashboard's scores, or its scoring code on the per-query artifacts (the first paragraph of this appendix), except the checks marked *: the raters follow a written guideline (the list-level raters see snippets only), the second judge reuses the production prompt, and the caselaw and finance audits use an LLM annotator. The machine-readable version is `paper/analysis/manifest.json`.

$-.010$, query-resampled 95% interval $-.024$ to $+.005$).

Vendor format change. From the hourly news run of 02:56 UTC on 30 October 2026, Google results fetched through Serper, and Jina results, which draw on Google, carry `https://www.google.com/goto?url=...` redirect links in place of the result URL, on 36 to 41 of every 100 news results in each later run that day (`goto_onset.py`). The title and snippet stay intact, so the news judge is unaffected, but the URL-based scholar, legal and filing matchers score such results as misses. The window of Table 1 ends on 28 October and contains no such URL. In the paired finance run of 30 October (Appendix H), Context returned such links on 15% of its results (`paired_analysis.py`).

Licenses and release. The rare stream is the `agentic/queries.parquet` artifact of the results dataset, sampled from three published agent-trajectory datasets: the DeepResearchGym search logs of Ning et al. (2026) (`cx-cmu/deepresearchgym-agentic-search-logs`, CC BY 4.0; 59% of sampled queries), the OpenResearcher trajectories of Li et al. (2026) (`OpenResearcher/OpenResearcher-Dataset`, MIT; 39%) and the LRAT-Train pairs of Zhou et al. (2026) (`Yuqi-Zhou/LRAT-Train`, no license stated on its card as of 30 October 2026; 1%). The repository describes the stream's filter (English-only language identification; rare-word test; removal of queries with a link, a host name under eight common top-level domains, or a hash, wallet or vehicle identifier; near-duplicate cap). The sources' terms govern the query texts; our code is MIT, and our CC BY 4.0 covers only our results, judgments and metadata. The next artifact revision drops the LRAT-derived queries unless their authors grant a license. These queries are 113 of the 6,901 filtered queries and 17 of the 1,200 rare query-runs of Table 1, and dropping them changes no engine's mean rare nDCG by more than .003 (`rare_drop_lrat.py`). A regular-expression scan in the same script of the 1,100 rare queries released with the window's runs finds no e-mail address and no `http` or `www` link; 66 contain a host-like name (61 in a `site:` operator) and 4 hold a token of 20 or more characters that mixes upper case, lower case and digits, such as an opaque identifier. Queries also name people, as the public source documents do, so a human reads the queries of each new artifact revision before release.

Removal requests go to the repository issue tracker. The per-query artifacts also redistribute the titles, URLs and snippets the APIs returned; we publish them as evaluation records under the vendors' terms as we read them and withdraw an engine's archived outputs on its request.

B Judge profile and gates

The news and rare judge runs in two steps, both with the same model. A *profile* call sees the query and today's date and returns: query type (*broad*, *specific*, *list*); domain (health, legal, financial, academic, tech, news, commerce, jobs, other); whether the query carries operators; one to three MAIN aspects, phrased as what a document must contain and never as the answer; auxiliary aspects; and a content archetype (*ephemeral*, *persistent*, *evergreen*) that sets the staleness rule. The profile is stored with the query and pasted verbatim into every per-result call, so the judge grades all engines against one interpretation of the query.

The *judgment* call sees the profile, today's date and one result (title, URL, published date if the API returned one, snippet capped at 2,000 characters) and emits a YAML block with eight gates and a rating. Gates: integrity (pass, limit, fail), relevance (entity, jurisdiction, topic), coverage of the MAIN aspects, completeness (list queries only), current-event freshness (dated events only), staleness by archetype, a domain cap (health, legal, financial, commerce and others carry hard caps), and operator compliance (quoted spans must appear verbatim in the snippet or title; `site:` must match the URL host). A failed relevance or operator gate forces 0; a failed coverage or completeness gate caps the rating at 2; staleness lowers the rating by one or two levels, or to 0 for a wrong year or entity status; a dated-event query whose result covers the entity but not the event is capped at 1. The rating maps to the Needs Met labels as 0 and 1 Fails to Meet (FailsM), 2 Slightly Meets (SM), 3 Highly Meets (HM) and 4 Fully Meets (FullyM). The judge is told to score facts against the page's own text and authority, never against its own knowledge of the answer. The prompts are in the repository under `src/needle/shared/prompts`.

C Operator mapping and configuration

Table 4 fixes the configuration behind every engine in Table 1: the endpoint, the variant or model where the vendor exposes one, and the request parameters that differ from the vendor default. We ask every engine for k results ($k = 10$ on scholar, 5 elsewhere) at its default tier and score a shorter list as returned.

Every query is parsed for the operators `site:`, `after:` and `before:` (ISO dates). Table 5 shows what each engine receives. Google (Serper), Bing (SearchAPI) and Jina get the raw string because the underlying engines parse these operators themselves. Quoted phrases are always passed through in the query text. In the 24 finance runs of Appendix F and the 10 legal runs of 18–28 October 2026, an operator or quoted phrase appears in 335 of 667 money, 333 of 445 filing, 64 of 128 caselaw and 200 of 400 CFR query-runs; 100 of the CFR query-runs quote the span (`ops_analysis.py`).

D Leak and quality filters

Scholar. A title query needs at least four words and a specific title (a digit, acronym, hyphenated term or five content tokens). A body keyword query needs at least three words and at least two content tokens absent from the title and abstract; a body question needs eight words and the same novelty test. A tip-of-tongue query needs twelve words, no quotes, no decimal or five-digit number, at most two content tokens shared with the title, and no coined term (acronym, camel case, digit-bearing word) that appears in the title or abstract. A paper enters only if all four queries pass. An instrumented run of the generator on 30 October 2026 passed 323 of 572 candidate papers (56%), from .41 in life sciences to .73 in health sciences (`scholar_rejects.py`). Each domain-by-age cell takes its first seven accepted papers, and nearly every cell of the 12 scholar runs of Table 1 holds 6 to 8, so the filter changes which papers a cell holds, not how many; whether the rejected papers would rank engines differently is untested.
Legal. A CFR query needs three words, no citation form (*CFR*, *U.S.C.*, §, "section n", "title n"), no part or section number of the gold section, and exactly one span of two to eight words that occurs verbatim in the section text; the quoted variant keeps the span in quotes, the others strip them.
News. The prompt that turns a news item into a query refuses evergreen items and returns a 2–6 word query on the specific event; query ids hash the text and the hour. Table 6 gives one generated query per bench and bucket.

Engine	Endpoint	Variant or non-default parameters
Keenable	api.keenable.ai	default; `keenable-realtime` is the realtime variant
Exa	api.exa.ai	`type=auto`; `exa-instant` and `exa-lite` are the instant and lite types
Google	google.serper.dev (Serper)	default
Bing	searchapi.io (SearchAPI)	engine `bing`
Brave	api.search.brave.com	web search; `brave-llmcontext` is the LLM-context endpoint
Parallel	api.parallel.ai	default processor; `parallel-turbo` is the turbo processor
Perplexity	api.perplexity.ai	search API; `perplexity-fast` is the fast mode
Tavily	api.tavily.com	default
Linkup	api.linkup.so	default
TinyFish	api.search.tinyfish.ai	default
Octen	api.octen.ai	default
Ceramic	api.ceramic.ai	default
You.com	ydc-index.io	default
Firecrawl	api.firecrawl.dev	search endpoint, default
Kagi	kagi.com/api/v1	default
Jina	s.jina.ai	default
Context	api.context.dev/v1	default
DeepSeek	api.deepseek.com	search tool endpoint (`/v1/tools/search`), default
Claude search	api.anthropic.com	`claude-sonnet-5` with the `web_search` tool
ChatGPT search	api.openai.com	`gpt-5.5` with `web_search`, `search_context_size=low`

Table 4: Configuration of the Table 1 engines. Operator handling per engine is in Table 5; the clients are in `src/needle/shared/search/`.

Engine	`site:`	`after:`/`before:`
Brave, Brave (LLM ctx)	in query text	freshness range
Keenable	in query text	published-date tokens
Exa	includeDomains	start/endPublishedDate
Tavily, Octen, TinyFish	include_domains	start/end dates
Linkup	includeDomains	from/toDate
Context	includeDomains	freshness (after only)
Perplexity	domain filter	date filters
Parallel	include_domains	after_date only
Kagi	lens sites	after/before filters
You.com	include_domains	freshness range
Firecrawl	in query text	cd_min/cd_max
DeepSeek	site: OR site:	appended to text
ChatGPT, Claude	allowed_domains	in prompt text
Ceramic	dropped	dropped
Google, Bing, Jina	raw string	raw string

Table 5: Ceramic drops the `site:`, `after:` and `before:` operators; Google, Bing and Jina get the raw string; the other engines get filter parameters or query text.

E Worked example of the ceiling

The gains are a design choice, top-heavy so that one FullyM result outweighs one HM result by half and an SM result by a factor of about 8.5; Appendix G shows the engine order under linear and binary alternatives. For engine e and query q with ratings $r_1, \ldots, r_5$,

$$\mathrm{DCG}_e(q) = \sum_{i \le 5} \frac{g(r_i)}{\log_2(i+1)},$$

$$\mathrm{nDCG}_e(q) = \frac{\mathrm{DCG}_e(q)}{\mathrm{IDCG}(q)},$$

with gains $g(4) = 1$, $g(3) = 0.667$, $g(2) = 0.117$, $g(1) = g(0) = 0$, and $\mathrm{IDCG}(q)$ the DCG of the ceiling's list. A bench score $M_b(e)$ is the mean over scored query-runs in the window (a failure scores 0 unless it is account-side, which is left out; on news and rare, every failure in the window scores 0). The ceiling's score $M_b(\mathrm{ceiling})$ is the same mean over the ceiling's lists (1 for news and rare), and the overall share is

$$S(e) = \frac{1}{|B|} \sum_{b \in B} \frac{M_b(e)}{M_b(\mathrm{ceiling})}$$

over the benches B the engine ran. For example, two engines answer one query. Engine A returns u_1 (rated 4), u_2 (2) and u_1 again under a tracking parameter (rated 3, but URL normalization strips the parameter, so the repeat scores 0).

Bench	Query	Gold or profile
news	Paramount Warner Bros takeover closes 2026	MAIN: the expected or actual closing date of Paramount's takeover of Warner Bros. in 2026; archetype *ephemeral*
finance, registry	nvidia number of employees	42,000, ±15%
finance, 10-Q	"servicenow" q3 fiscal 2024 operating income	$418M, ±0.1%
finance, filing	Huntington Bancshares Ann B. Crane director vote 2026 after:2026-03-10 before:2026-06-08	accession 0000049196-26-000038
scholar, title	optimally selling heterogeneous objects by quantity	arXiv 2610.36244
scholar, body keyword	gap rent regularity	same paper
scholar, body question	Which Go City pass for New York does the paper cite as a quantity-menu example, and what restriction applies to each selected attraction?	same paper
scholar, tip-of-tongue	Wasn't there a paper about a seller offering different goods at prices based on how many a buyer takes, letting them choose which ones, and showing this can bring the most revenue under certain value patterns?	same paper
legal, CFR	"modify or waive any" county committees authority	7 CFR 1417.2
legal, caselaw	citation insurance v broan-nutone first circuit after:2026-07-14 before:2026-10-12	docket 25-1426 next to a party token; CourtListener id 10961606
rare	"Krasnogvardeysky District" Russia	judged; rare word splits into 7 pieces

Table 6: News and rare queries carry no gold: the judge profile fixes what a result must contain. One generated query per bench and bucket, with the gold or judge profile it is scored against.

Engine B returns u_3 (3), u_1 (4) and u_4 (1). The *ceiling* keeps the best rating per normalized URL, $\{u_1 : 4, u_3 : 3, u_2 : 2, u_4 : 1\}$, in rating order, so IDCG $= 1 + 0.667/1.585 + 0.117/2 = 1.479$. Engine A scores $(1 + 0.117/1.585)/1.479 = 0.73$ and Engine B $(0.667 + 1/1.585)/1.479 = 0.88$. We score an engine that returns fewer than k results on the list it returned, against the ceiling's list, which still takes up to k results from the pool. Had no engine returned a result rated 2 or higher, IDCG would be 0 and the query would be dropped from every engine's mean. Ties in the pool do not matter, because only ratings enter IDCG. The duplicate penalty does not apply to a query that carries `site:`; no news query in the window carried `site:`, and on the rare bench the exception changes 3 of 1,748 `site:` engine–query pairs. Because the ceiling keeps each URL once, such a repeat can lift an engine's nDCG@5 above 1: Tavily's list for one rare query on 1 October, outside the Table 1 window, scores 1.34.

F Finance matcher check

The finance matcher accepts a money figure within 0.1% of gold and a headcount within 15%, and its LLM backstop may upgrade a non-money miss (Appendix A). We checked the bands and the backstop on the 24 daily finance runs from 29 September to 28 October 2026, a longer span than the 10 runs of Table 1. These runs cover 21 engines and 56,760 engine–query pairs without an account-side failure, of which 13,149 ask for a quarterly money figure and 4,393 for a headcount (`band_sweep.py`).

Tolerance sweep. Table 7 scores the stored results under other bands (`band_sweep.py`). A 2% money band would add 9 recall points to the 0.1% band, and the 15% headcount band adds 10 points over 1%. Counting individual matched results, 21.7% of the money matches within 2% lie outside 0.1%, and 35.6% of the headcount matches within 15% lie outside 1%.

Money band	0	**.1%**	.5%	1%	2%	5%	10%
recall@5	.363	**.447**	.498	.517	.541	.583	.628
τ_b vs. .1%	.97	1	.95	.94	.98	.91	.88
Headcount band	0	1%	5%	**15%**	25%	50%	
recall@5	.794	.811	.863	**.915**	.919	.935	
τ_b vs. 15%	.73	.77	.81	1	.93	.79	

Table 7: Wider bands raise recall at every step. The bold bands are the matcher's. The engine order barely moves across money bands up to 2% ($\tau_b \geq .94$) and moves more across headcount bands. Deterministic recall@5 without backstop or table reading over all engine–query pairs of the 24 audit runs, and Kendall's τ_b of the 21 engines' order against the bold band (1 means the same order), by tolerance.

Precision audit. On these runs, an LLM backstop on every field behind a 2% money band judged 12,919 engine–query pairs that band missed and upgraded 3,187 (24.7%); 2,621 of the upgrades are money queries (49% of money misses), against 1.8% of identifier misses. We sampled 399 results: 80 money results 0.1–2% from the gold, 140 money hits within 0.1% (the *exact-band* stratum: 40 in a first sample and 100 more drawn from the Table 1 window), 40 headcount hits that depend on the band (1–15%) and 10 within 1%, 59 hits on the other field types in proportion to their frequency, and 70 backstop upgrades (50 money, 20 other). One LLM annotator per item (Claude Opus 5.5) judged whether the result states the gold value, with reporting rounding allowed and a value for a different period or line item counted as wrong. A first pass read 700 characters per result. The 58 hits it rejected were judged again on the full 2,000-character text the matcher reads, and 21 reversed (`audit_recheck.py`). The 33 rejected backstop upgrades were not judged again, so the upgrade rates below may be low. Money hits within 0.1% are right at .93, banded headcount hits at .93 and other-field hits at .98; all 10 headcount hits within 1% hold. Two strata are weak, and the matcher accepts neither: money results 0.1–2% from the gold (.71) and money upgrades of the backstop, which is lenient on money (.36) and sound elsewhere (.95 on 20). Weighted by hit frequency, the matcher without its table reading has precision .98 overall, .93 on money, .97 on headcounts and .98 on the other fields; a 2% money band would lower its money precision to .88 and its overall precision to .97 (95% Wilson intervals: money within 0.1% .87–.96, money 0.1–2% from the gold .61–.80, headcount banded .80–.97, other fields .91–1.00). Most money false positives within 0.1% are a figure whose period or line item the snippet leaves open (a table without column headers, an EPS figure not marked as diluted); most of those 0.1–2% from the gold are a neighboring quarter or a related line item, such as adjusted net income or basic EPS (`aggregate_audit.py`).

Table reading. The matcher also reads unit-less table figures and parenthesized losses (Appendix A). We sampled 80 results whose unit-less figure or parenthesized loss equals the gold in thousands or millions: 73 are confirmed (.91, Wilson .83–.96), and four of the seven others are tables that stop before the queried row, two a different period. The table reading, which scales a bare figure only after a money word, matches 73 of the 80, and 71 of them are confirmed. On the 24 audit runs, the non-money backstop supplies at most 1.4% of finance hits (`aggregate_audit.py`). A figure within 0.1% can still be a neighboring quarter's identical value, so money is the matcher's weak spot.

False negatives. We also sampled 100 money query-runs, across engines and runs, that the matcher without its table reading misses even with a 2% money band, and asked an LLM annotator (Claude Opus 5.5) whether any of the five results states the gold value for the right company, period and metric. The annotator found the value in 29 (95% interval .21–.39) and a near miss (a different period or line item) in 2 more; 20 of the 29 were at rank 1 or 2. Nearly all of the 29 use formats this matcher does not read: bare table figures in millions or thousands without a unit word ("801" for $801M), losses in parentheses ("(458)") and losses stated in words ("Operating loss was $56.3 million"). The table reading recovers 19 of these 29 and matches 1 of the 71 others. Of the 80 that stay misses, 10 (.125) still hold the answer, and a money result 0.1–2% from the gold states it .71 of the time (precision audit). Against this, weighted by hit frequency, .07 of exact-band money hits, .03 of headcount hits and .02 of other-field hits are false positives, and so are 2 of the 73 table-reading hits (.03). We apply these rates by stratum (`finance_bias_table1.py`, on the 10 finance runs and 21 engines of Table 1, with account-side failures and the queries the scorer drops after judge errors excluded; money queries are 23% of the window's finance queries, headcount 8% and the other fields 69%). For an engine with field shares p_m, p_h, p_o, per-field recall r_m, r_h, r_o, share x_m of its money queries that only the table reading matches, share a_m of its money queries with at least one result and share b_m that miss with a result 0.1–2% from the gold (367 of the window's 2,282 money misses with results), expected false positives are $p_m[.07(r_m - x_m) + .03x_m] + .03\,p_h r_h + .02\,p_o r_o$ and expected false negatives $p_m[.71\,b_m + .125(a_m - r_m - b_m)]$. The net correction runs from $-.025$ for Keenable, $-.020$ for Parallel and $-.017$ for Exa to $+.015$ for Claude search and $+.021$ for Ceramic. False hits and missed answers therefore do not cancel: reported finance recall runs .002 to .025 high for the

nine engines above .80 and .001 to .021 low for the twelve below. Correcting every engine compresses the finance spread from .481 to .443, puts Perplexity (reported .895) just above Keenable (.903), .879 against .878 after correction, and ChatGPT search (.809) above DeepSeek (.818), .808 against .803. With the corrected recall over the reported ceiling as the finance share, the overall order keeps τ_b .99: Kagi passes ChatGPT search, and Keenable leads Exa by 4.5 points. The miss sample is too small to show whether the error rates differ by engine.

Fresh window. To validate the matcher on unseen data, we generated finance queries with the production generator on 30 October 2026, kept all 146 money queries (net income, operating income, diluted EPS) and a stratified 161 of the other fields, and sent the 307 queries to 17 configurations between 00:47 and 01:38 UTC on 31 October (`fresh_finance.py`, `audit/fresh/`). Claude search was left out because the account's credit had run out. Google and Jina carried the redirect URLs of Appendix A, which hide the host but not the text the money and name matchers read. One LLM annotator per item (Claude Opus 5.5, in 10 sessions of up to 50 items) read at least the matcher's full 2,000-character text and graded every table-reading hit but one (335), 100 exact-band money hits and 50 other-field hits. Each graded item is the first matching result in a query's top five, so the precision below is that of query-level recall@5 decisions. Exact-band precision is .93 (93 yes, 6 unclear, 1 no; `fresh_finance_labels.py`), as in the precision audit. The table reading is weaker and engine-specific. Of its 335 graded hits (`fresh_pool_diff.py`), 235 state the gold figure for the right period (.70, Wilson .65–.75), 95 show the figure in a table whose period columns the snippet does not label, and 5 contradict it. We treat the 95 as unresolved: precision is .70 if an unresolved hit counts as wrong and .99 if only contradictions do. Confirmed shares differ by engine: Firecrawl 13 of 43, Tavily 12 of 24, ChatGPT search 18 of 27, Keenable 14 of 20, Brave (LLM ctx) 29 of 39 and Parallel 36 of 48, against Perplexity 27 of 31 and Exa 42 of 44. Of the 50 other-field hits, 39 are confirmed; of the other 11, 4 are `website` hits matched on the URL host, which the guideline told the annotators to ignore, and 2 are headcounts inside the 15% band, so the rule-consistent figure is .90 (`fresh_finance_labels.py`). To bound the effect on Table 1, we multiply each engine's table-reading hits in the window by its fresh unconfirmed share (its own where 20 or more hits were audited, .30 otherwise) and subtract the result from finance recall (`reading_correction.py`). Firecrawl loses 6.0 points, Tavily 3.2, Parallel 2.3, DeepSeek 2.2, Brave (LLM ctx) 2.1, ChatGPT search 2.1 and Keenable 1.3; every other engine loses 1.0 or less. The finance order keeps τ_b .90 and the overall order .98: ranks 1 to 10 do not move, and ChatGPT search–Kagi and Firecrawl–Octen swap. A period-aware table reading is future work. Dropping the table reading (`no_table_reading.py`) lowers finance recall by 0.1 to 7.5 points (Firecrawl 7.5, Parallel 6.8, Tavily 5.7, Exa 5.6, ChatGPT search 5.5, Brave (LLM ctx) 5.4, DeepSeek and Perplexity 4.6, the rest under 2) and the finance ceiling from .993 to .988; the finance order keeps τ_b .84 and the overall order .97, ranks 1 to 3 are unchanged, and Parallel–Brave (LLM ctx), ChatGPT search–Kagi and Firecrawl–Octen swap. As a check of the finance column, the fresh run's engine shares agree with the Table 1 finance shares at τ_b .80 with a mean absolute difference of .05 (`audit/fresh/summary.json`), and the same five engines lead (Exa .96, Keenable .95, Parallel .95, Brave (LLM ctx) .94, Perplexity .94, against .86–.93 in the table). Half its queries are money queries, against 23% in production; it covers one day, has no LLM backstop for the non-money fields and ran 17 of the 21 configurations, so it checks the direction of the finance column, not its values.

G Robustness checks

The appendices use every active configuration, so names absent from Table 1, such as Parallel Turbo or Exa Instant, appear here. Table 9 collects the alternative views of the engine order from this appendix and Appendix F.

Leave-one-engine-out ceiling. On 230 of the 232 news runs of Table 1, 17–28 October 2026 (4,516 query-runs with a ceiling; `loo_ceiling.py`, `audit/news_check_runs.json`), we recomputed the nDCG@5 of each of the 25 configurations from the per-query artifacts with the dashboard's scoring code (account-side failures left out) against a ceiling pooled from the other 24 configurations only; a variant of the same engine

stays in the pool. Because every returned result is judged, the ceiling can only fall when an engine leaves the pool, so the left-out engine's share can only rise. The largest change is +.011 (DeepSeek). Every other configuration moves by .005 or less. DeepSeek passes both Perplexity configurations and Exa passes both Keenable configurations; the rest of the order is unchanged (Kendall $\tau_b = .97$). An engine beats its leave-one-out ceiling on at most 1% of queries. With a ceiling pooled from the 21 engines of Table 1 only, the same check for every bench (`loo_all.py`) moves no scholar, finance or legal share by more than .007, no news share by more than .012 and no rare share by more than .037 (ChatGPT search). The overall share moves by at most 0.8 points, and only Context and DeepSeek swap in the overall order. A judged query whose pool the removal empties is dropped for that engine (11 engine–query cases over 4,515 news and 22 over 1,042 rare query-runs).

Empty ceilings. A judged query that no engine answers with a rating of 2 or more has an empty ceiling and is dropped: 6 of the 4,522 fully judged news query-runs in 230 of the window's 232 news runs (0.1%) and 147 of the 1,189 fully judged rare query-runs (12.4%; 129 of the 499 quoted and 18 of the 690 unquoted). A query-run with a failed judgement is dropped as well (59 of the 4,581 news and 11 of the 1,200 rare query-runs). Scoring the empty-ceiling query-runs as zero for every engine and for the ceiling instead lowers every absolute mean nDCG in proportion but changes no bench share by more than .002 and swaps only Octen and Claude search on rare (`ceiling_drop.py`), because the zeros enter the engine mean and the ceiling mean alike. The absolute nDCG columns of Table 1 are therefore conditional means.

Sensitivity to scoring choices. On the 230 news runs of the leave-one-out check we recomputed the news shares of the 25 configurations under other gain mappings and compared their orders with the default by Kendall τ_b (`sensitivity.py`, `audit/sensitivity.json`): linear gains $\{1, .75, .5, .25, 0\}$ and the exponential mapping $(2^r - 1)/15$ both give .95 (no configuration moves more than two places), binary relevance at rating ≥ 3 gives .98, at $= 4$ .81 (Octen rises five places), and at ≥ 2 .87 (Exa and Parallel rise five). Ranking Table 1 by the mean of the five absolute metrics instead of share of ceiling gives $\tau_b = .99$; Context and DeepSeek swap 8th and 9th.

Paired differences. In Table 1, 15 of the 20 adjacent pairs have overlapping marginal intervals. Overlap of two marginal intervals is neither a test of equivalence nor transitive, so we also ran a paired test: in each run resample we recomputed every engine's share and took the difference of each pair (`adjacent_pairs.py`, 2,000 resamples, Table 1 runs). The paired 95% interval of the difference excludes zero for 9 of the 20 adjacent pairs and for 188 of the 210 pairs overall. With a Bonferroni adjustment for 210 comparisons (normal approximation on the bootstrap standard deviation, $z = 3.68$) 178 pairs stay separated. The 32 adjusted-unresolved pairs form three groups: Exa to Brave (LLM ctx) (Exa–Perplexity $+1.6$, Perplexity–Parallel $+2.0$, Perplexity–Brave (LLM ctx) $+2.3$, Parallel–Brave (LLM ctx) $+0.3$), Linkup to You.com (Linkup, Brave, Context, DeepSeek, You.com), and ChatGPT search to Tavily (ChatGPT search, Kagi, Firecrawl, Octen, Google, Claude search, TinyFish, Bing, Jina, Tavily; Octen–Bing $+6.1$, Google–Bing $+3.6$ and Google–Claude search $+2.6$ are its widest links); only Keenable and Ceramic are separated from every other engine (`audit/adjacent_pairs.json` lists every pair). Resampling days instead of runs, so that the hourly news runs of one day move together, separates 7 of the 20 adjacent pairs, 185 of 210 pointwise and 174 adjusted (`audit/adjacent_pairs_day.json`).

Repeated queries and cluster resampling. Runs are not fully independent units. Counting every query-run whose exact text appears in another run of the window (first occurrences included), 33% of finance query-runs recur (registry facts), 16% of rare, 11% of news (mostly within one day), 5% of scholar and 2% of legal; counting only occurrences after the first, the shares are 21%, 8%, 6%, 3% and 1% (`repeat_queries.py`). Day-level resampling leaves unmodeled the cross-day repetition of 33% of finance, 16% of rare and 4% of news query-runs. Resampling clusters of identical query text instead of runs (for scholar, the paper behind the four queries; `cluster_pairs.py`) separates 188 of the 210 pairs pointwise and 180 after the Bonferroni adjustment, with Keenable–Exa at $+4.7$ ($+3.7$ to $+5.7$); its intervals are narrower than the run-level ones for 145 of the 210 pairs. The paper reports the run-level counts.

Engine	Share	95% CI	Δ matched	95% CI	Sch.	News	Fin.	Rare	Legal
Keenable	73.2	72.3–74.1	+1.6	+0.4, +2.7	.811	.609	.909	.508	.821
Exa	68.5	67.4–69.4	−1.1	−2.3, +0.2	.625	.599	.929	.476	.795
Perplexity	66.9	65.3–68.5	−0.7	−2.5, +1.1	.641	.640	.902	.468	.694
Parallel	64.9	63.6–66.1	−2.1	−3.5, −0.7	.651	.535	.914	.387	.757
Brave (LLM ctx)	64.6	63.4–65.6	+2.5	+1.1, +3.9	.591	.642	.865	.454	.678
Linkup	57.6	56.1–58.9	+1.7	−0.1, +3.8	.521	.567	.803	.400	.588
Brave	57.3	56.2–58.4	+1.8	+0.2, +3.2	.541	.557	.775	.354	.640
Context	57.1	55.4–58.6	−0.9	−3.0, +1.1	.664	.469	.788	.339	.594
DeepSeek	56.9	55.8–57.9	+1.1	−0.6, +2.9	.301	.635	.824	.387	.696
You.com	55.8	54.7–56.9	+1.6	+0.3, +2.9	.523	.553	.752	.343	.620
ChatGPT search	51.5	49.4–53.1	−0.5	−2.0, +0.8	.333	.340	.816	.366	.718
Kagi	51.3	49.9–52.6	+0.1	−0.8, +1.2	.621	.301	.734	.275	.635
Firecrawl	50.7	49.2–52.0	−0.5	−1.9, +0.9	.541	.286	.826	.265	.616
Octen	50.2	48.8–51.7	−1.1	−2.7, +0.6	.340	.511	.735	.322	.604
Google (Serper)	47.7	46.8–48.6	−0.3	−1.5, +0.9	.527	.262	.743	.255	.600
Claude search	45.1	42.7–47.6	+1.2	−0.4, +2.9	.553	.276	.690	.195	.541
TinyFish	44.5	43.0–45.9	−0.4	−2.0, +1.2	.476	.256	.736	.220	.537
Bing (SearchAPI)	44.1	41.1–46.8	+4.1	+1.5, +7.5	.461	.288	.680	.173	.604
Jina	43.8	42.4–45.1	−0.5	−2.2, +1.2	.545	.210	.731	.163	.539
Tavily	42.6	41.5–43.7	−0.1	−1.2, +1.3	.336	.303	.834	.284	.372
Ceramic	14.7	13.9–15.7	−0.1	−1.1, +1.2	.002	.040	.444	.109	.141

Table 8: Under matched query support six engines move with intervals that exclude zero, most Bing (+4.1), Brave (LLM ctx) (+2.5) and Parallel (−2.1). Columns give Table 1 in numbers: overall share (%) with its run-level bootstrap interval; the change in overall share under matched query support over the whole window (every engine scored on the 4,048 query-runs no engine lost to an account-side failure; Appendix G) with its run-resampled interval; and the bench shares. Both shares in the Δ matched column come from the per-query artifacts (`matched_pairs.py`); their base exceeds the Share column by up to 6.3 points (Claude search; Appendix A).

Failures. We classify each failed request from its own error record with the rule of Appendix A. A failure is account-side when it comes from our own account: an HTTP 401, 402, 403, 429, 432 or 433 response, or an HTTP 400 response or API error whose message names a credit balance, too few credits, a usage limit, a quota, billing or an API key; a rate limit counts as our quota. On finance, scholar and legal, Table 1 excludes account-side failures and scores every other failure (such as timeouts, server errors and invalid requests) as zero, which is what an agent experiences; on news and rare it scores every failure zero (Appendix A). Over all five benches of the Table 1 runs the account-side failures are Claude search's 4,423 refusals because our credit balance ran out (35.9% of its requests), TinyFish's 3,583 HTTP 429 rate limits (29.1%), ChatGPT search's 1,351 HTTP 429 responses saying that our credits had run out (11.0%), Kagi's 940 HTTP 401 responses (7.6%) and Jina's 304, Parallel's 100 and Exa's 40 HTTP 402 payment errors (2.5%, 0.8% and 0.3%). The dashboard excludes those on finance, scholar and legal: 27.9% of TinyFish's query-runs, 16.9% of Claude search's, 8.7% of ChatGPT search's, 4.7% of Kagi's and 2.5% of Jina's, and none of Parallel's or Exa's. Other failures are 2.4% of Jina's query-runs (HTTP 422 errors), 0.4% of Bing's and at most 0.1% of every other engine's. Each request is sent at most three times before a failure is recorded (Appendix A). The account-side failures cluster by date: ChatGPT search's fall on 17, 18, 22 and 27 October, Jina's on 20 October, Claude search's on 22–26 October, peaking on 23–25 October, Parallel's on 22 October, Kagi's on 27–28 October, Exa's on 19 October and TinyFish's on all 12 days. Claude search's, Kagi's, Exa's and Parallel's split evenly between the first and second half of the query list; ChatGPT search's (431 and 920), Jina's (96 and 208) and TinyFish's (1,657 and 1,926) fall more in the second half (`failure_profile.py`, `audit/failure_profile.json`). Scoring only answered requests for every engine instead moves Claude search from 45.1 to 51.4, rank 16 to 13. Every other engine moves by at most 0.9 points, and the answered-only order agrees with Table 1 at Kendall $\tau_b = .96$.

Matched support. Because the exclusions leave TinyFish, Claude search, ChatGPT search, Kagi and Jina scored on fewer query-runs than the other engines, we also apply a matched-support rule: a query-run is dropped for every engine when an account-side failure occurred on it, and other fail-

ures stay zeros. We applied the rule to the whole Table 1 window (230 of its 232 news runs and every other run; `matched_pairs.py`), scoring each query-run from the per-query artifacts with the dashboard's scoring code. On finance, scholar and legal this per-query base reproduces every dashboard row; on news and rare it excludes the account-side failures that those runs score zero (Appendix A), so it exceeds Table 1 by 6.3 points for Claude search (51.4 against 45.1), 0.6 to 0.9 for ChatGPT search, Kagi and TinyFish and at most 0.2 for every other engine. The rule retains 4,048 of the 12,313 query-runs in these runs (the full window has 12,353) and agrees with the order of Table 1 at $\tau_b = .89$. Six differences have run-resampled 95% intervals that exclude zero: Bing $+4.1$ ($+1.5$ to $+7.5$, rank 18 to 16), Brave (LLM ctx) $+2.5$ ($+1.1$ to $+3.9$, 5 to 3), Parallel -2.1 (-3.5 to -0.7, 4 to 5), Brave $+1.8$ ($+0.2$ to $+3.2$), Keenable $+1.6$ ($+0.4$ to $+2.7$) and You.com $+1.6$ ($+0.3$ to $+2.9$). Every other engine moves by at most 1.7 points with an interval that covers zero; Keenable and Exa stay first and second, and Tavily and Ceramic last. On matched support the paired test of *Paired differences* above separates 7 of 20 adjacent pairs, 189 of 210 pairs pointwise and 171 with the Bonferroni adjustment, with only Keenable and Ceramic separated from every other engine. Across the three estimates (run-level, day-level and matched support; `pairs_survive.py`) 164 of the 210 pairs stay separated under the adjustment in all of them. Fourteen adjusted primary separations fail at least one check: Claude search against ChatGPT search, Kagi, Firecrawl and Octen, and Firecrawl–Bing, fail day-level resampling and matched support; Exa–Brave (LLM ctx), Parallel–Linkup, Parallel–Brave, Linkup–You.com, Context against ChatGPT search, Kagi and Claude search, ChatGPT search–Bing and Google–TinyFish fail matched support (`audit/pairs_survive.json`). We keep the unequal-support estimate as the primary one because it uses all 12,353 query-runs rather than 4,048, and print the matched-support change beside it in Table 8.

Exclusion bounds. Account-side failures on finance, scholar and legal are excluded from the scores of TinyFish, Claude search, ChatGPT search, Kagi and Jina, while the Failed column of Table 1 still counts them. Exclusion assumes the engine would have scored on the failed requests as it did on the rest; two imputations bound the alternative (`exclusion_bounds.py`): scoring every excluded request zero or scoring it at the ceiling puts Claude search's overall share between 34.7 and 52.0 around the reported 45.1 (21–34% of its scholar, finance and legal requests excluded), TinyFish between 37.6 and 52.0 (44.5; 71.5% of its scholar requests excluded), Kagi between 47.5 and 53.2 (51.3), ChatGPT search between 48.2 and 54.5 (51.5) and Jina between 42.9 and 44.4 (43.8); no other engine has account-side exclusions in the Table 1 runs and so no range. The excluded fraction counts account-side failures and, on finance, the queries of the 19 October run that the scorer drops after judge errors, not the news and rare query-runs that every engine drops for an empty ceiling or a failed judgement. Without the assumption Claude search and TinyFish range over ranks 11 to 20, Kagi over 11 to 15, ChatGPT search over 11 to 14 and Jina over 18 to 19; ranks 1 to 10, and so the Keenable–Exa margin of 4.7 points, hold under either extreme.

Common support and uncertainty. Engines joined the benchmark at different dates, so Table 1 restricts every bench to the runs since 17 October, when the last engine joined, in which all 21 configurations ran. The only gaps are the query-runs excluded for account-side failures on finance, scholar and legal: 2.5% for Jina, 4.7% for Kagi, 8.7% for ChatGPT search, 16.9% for Claude search and 27.9% for TinyFish, plus the finance queries of the 19 October run that the scorer drops after judge errors. Kagi's whole-run failures in the news and rare runs of 27–28 October score zero; the dashboard has no Kagi score for later news and rare runs (Appendix A), so the window of those benches ends there. Over the eight weeks of Figure 1b, adjacent weekly orders of the listed engines agree at τ_b .81–.94 (mean .89), and the 15 configurations present in every week give .85–.94 (mean .91; `weekly_tau.py`, `audit/weekly_tau.json`); Keenable leads every week. Pooling the news ceiling over those 15 engines only, on the sampled runs of weeks 40–43, raises their weekly news shares by .003 to .010 on average and changes no weekly order ($\tau_b = 1$; `fixed_pool_weekly.py`), so the growing pool does not drive the trends in Figure 1b. Claude search's week-43 share in Figure 1b (44.0) includes our credit lapse, which began at 08:15 UTC on 22 October: the dash-

board excludes it on scholar, finance and legal but scores it zero on news and rare. If each week keeps only the runs in which Claude search lost at most 10% of its requests, its share falls from 58.3 in week 39 to 52.2 in week 43, mostly by week 41, when its failures were below 8%; the rest of the figure's fall from 56.6 is the lapse. Scoring every failure as zero gives 54.6 in week 39 and 24.7 in week 43 (`claude_weekly.py`, `audit/claude_weekly.json`).

Where the scholar lead comes from. Of Keenable's 4.7-point lead over Exa in overall share, scholar contributes 3.7 points, rare 0.6, legal 0.5, news 0.2 and finance -0.4 (`bench_weights.py`). Dropping any one bench keeps Keenable first (τ_b .86–.93 against Table 1) but shrinks the lead to 1.2 points without scholar. Split by stratum over the 12 Table 1 scholar runs (4,820 query-runs; `scholar_strata.py`), Keenable's recall@10 exceeds Exa's by $+.38$ on papers 7 days old (.79 against .41), $+.10$ at 30 days and $+.03$ at one year; by $+.20$ to $+.30$ in computer, life, physical and social sciences but by $-.10$ in health sciences; and by $+.01$ on title queries, $+.17$ on body keywords and tip-of-tongue descriptions and $+.33$ on body questions. The lead is therefore a freshness and full-text lead on recent non-medical papers, not a uniform one.

Known-item hits are the target page. A matcher hit shows that a returned result carries the gold identifier, not that it is the target document. On the three scholar runs of 25–28 October (13,311 hits) the identifier appears in the hit's own URL for 95.1% of hits, and 84% of hits are an arXiv page. On the ten downloaded legal runs, the section number appears in the URL of 94.0% of CFR hits and 95.4% sit on a primary legal host (eCFR, LII, GovInfo). For caselaw, 62.2% of hits carry the docket, cluster id or citation in the URL and 86.8% sit on a primary host (CourtListener, Justia, supremecourt.gov, uscourts.gov). An LLM annotator (Claude Opus 5.5, engine names hidden) classified 100 caselaw hits drawn four per configuration from the ten runs: 80 are the opinion or a court docket page of the gold case, 14 are pages about the same case (Wikipedia, news summaries, SCOTUSblog, a motion in the same docket), 5 are unclear and 1 is a page about another case. Caselaw recall thus credits a page about the right case rather than the target in 14 to 19 of 100 hits. Scaling every engine's caselaw hits by the pooled target fraction .80 (Wilson .71–.87) lowers legal recall by .01 to .04 and keeps the legal order at τ_b .96 (`legal_target_only.py`); applying each configuration's own sampled fraction, which four hits bound only to within .25, gives τ_b .78, so the audit rules out a large pooled effect but not engine-specific ones. In the other direction, the annotator checked 100 scholar body-query runs that the matcher scored as misses: the target paper appears among the ten results, under a URL or snippet that carries no identifier, in 11 (95% interval .06–.19; project pages, aggregators, a podcast page, a news summary), never as a citation inside another document. Scholar recall is thus a lower bound.

A later day, all benches. The production benches kept running after the Table 1 window, and their archived runs give a one-day check outside it: from the 01:17 UTC news run of 30 October to the finance run of 03:08 UTC on 31 October, 28 news runs and one scholar, rare, legal and finance run each (1,224 query-runs per engine), as the dashboard publishes them; its news and rare runs leave out every failure (`later_day.py`, `audit/later_day.json`). Nineteen configurations have all five benches: Claude search has no finance run because the account's credit had lapsed, and every Kagi request failed. The overall order agrees with Table 1 at τ_b .86 (per bench .71 to .89): the same five engines lead and Ceramic is last. The leader changes: Perplexity 72.4 and Keenable 71.3, then Parallel 65.8, Exa 63.7 and Brave (LLM ctx) 61.7. The change comes from the single scholar run (Perplexity's bench share .64 to .77, Keenable's .81 to .77) and the single legal run of 40 queries (Keenable .82 to .73, Perplexity .69 to .78); both moves lie within the run-to-run spread of those benches (per-run bench shares in the runs of Table 1: scholar, Keenable .74–.90 and Perplexity .56–.78; legal, Keenable .71–.89 and Perplexity .47–.85). Finance rises for every leader (.89 to .96), and news holds. Google and Jina returned redirect URLs for 51% and 50% of their results that day and fall 4.7 and 6.8 points. One day is a direction check, not a new leaderboard.

Blinded raters with page text. To test the judge against an independent reading, we sampled 100 news and 100 rare queries from the 17–29 October runs with at least 19 engines and drew six distinct top-5 URLs per query (1,200 results, judge ratings 0:441, 1:120, 2:263, 3:226, 4:150). We fetched each page (1,130 fetched; the rest are login walls,

Alternative view	What moves	Order vs. Table 1	Leader and lead over Exa
Matched query support (4,048 query-runs)	Bing $+4.1$, Brave (LLM ctx) $+2.5$, Parallel -2.1, Brave $+1.8$, Keenable and You.com $+1.6$; others within ± 1.7	τ_b .89; 171 of 210 pairs separated (adjusted)	Keenable, $+7.2$
Day-level resampling	intervals widen	174 of 210 pairs separated (adjusted)	Keenable, $+4.7$
Query-cluster resampling (identical text; scholar: paper)	most intervals narrow	180 of 210 pairs separated (adjusted)	Keenable, $+4.7$
Excluded requests scored zero or at the ceiling	Claude search 34.7–52.0, TinyFish 37.6–52.0, Kagi 47.5–53.2, ChatGPT search 48.2–54.5	Claude search and TinyFish ranks 11–20, Kagi 11–15, ChatGPT search 11–14; ranks 1–10 unchanged	Keenable, $+4.7$
Ceiling pooled without the scored engine	overall share ≤ 0.8 points; rare share $\leq .037$	Context and DeepSeek swap	Keenable, $+4.2$
One bench dropped (each of five)	lead from $+1.2$ (no scholar) to $+6.4$ (no finance)	τ_b .86–.93	Keenable in all five
Rare column from unquoted queries only	rare: Exa, Perplexity, Brave (LLM ctx) lead; Keenable fourth	overall τ_b .96	Keenable, $+3.3$
Snippets cut to 200 characters (8 news queries)	news shares move .084 on average	news τ_b .73 vs. the second judge's full-snippet order	leader and bottom two unchanged
Finance error corrected by stratum	finance recall -2.5 (Keenable) to $+2.1$ (Ceramic); finance spread .48 to .44	on finance, Perplexity passes Keenable and ChatGPT search passes DeepSeek; overall τ_b .99, Kagi passes ChatGPT search	Keenable, $+4.5$
Finance without the table reading (Appendix F)	finance recall -0.1 to -7.5 (Firecrawl, Parallel, Tavily, Exa most)	finance τ_b .84; overall τ_b .97; ranks 1–3 unchanged, three adjacent swaps	Keenable, $+5.5$
Fresh finance run, 31 Oct 2026 (307 queries, 17 engines; finance column only; Appendix F)	finance shares move .05 on average	finance τ_b .80; same five engines lead	Exa .96, Keenable .95 on finance
Later day, all benches (30 Oct 2026; 19 engines)	one scholar and one legal run move the leaders	overall τ_b .86; same five lead, Ceramic last	Perplexity 72.4, Keenable 71.3
Rater grades instead of the judge (100 complete pools, Gemini 4 and Grok 4.7; not a full overall score)	engine nDCG shares move .06–.07 on average	Spearman .94 vs. the judge (τ_b .83–.85)	same five engines lead and same three trail; Keenable–Exa unresolved under the judge and both raters
Empty-ceiling queries scored zero	absolute news and rare means fall 0.1% and 12.4%	shares move $\leq .002$; one rare swap	Keenable, $+4.7$

Table 9: Every retrospective view that recomputes the full overall score keeps Keenable first and Ceramic last, while the later day of 30 October puts Perplexity first by 1.1 points. The rater-grade check preserves the top group and the bottom three but not the order within the top group; the 200-character check preserves the leader and the bottom two but not the top group. Rows give the alternative views analyzed in this paper; columns give what each view moves, its order against Table 1 or the named reference, and the leader with its lead over Exa.

blocked hosts or dead pages) and gave two Claude Opus 5.5 raters (A and B) a written guideline (0–4 Needs Met scale with the same operator rules, in `analysis/audit/raters/`) and, per item, the date of the run, the query, the URL, the title, the snippet and the first 3,000 characters of the page, with the engine hidden. Each rater saw the 1,200 items in its own random order in 24 sessions of 50 and recorded a snippet grade and a page grade per item. The 128 items with a missing page are the 70 pages no fetch returned plus 58 fetched pages whose text a rater marked as broken. Gemini 4 Argon and Grok 4.7 graded the same items under the same guideline, one item per call at temperature 0, through OpenRouter (`third_rater.py`, `third_rater_analysis.py`). Table 11 gives the agreement. The two Claude raters agree at quadratic κ .94 (query-resampled 95% CI .93–.95), Gemini and Grok agree with each other at .92, and they agree with the judge at

Engine	title	body keyword	body question	tip of tongue	query-runs
ceiling (pooled)	*1*	*.888*	*.938*	*.767*	*12,020*
Parallel	**.981**	.453	.642	.278	12,020
Kagi	.962	.674	.640	.091	11,599
Bing (SearchAPI)	.903	.402	.411	.060	12,020
Parallel Turbo	.894	.208	.294	.110	12,020
Brave (LLM ctx)	.893	.440	.620	.194	12,020
ChatGPT search	.886	.083	.192	.118	11,016
You.com	.883	.370	.593	.129	12,020
Brave	.881	.383	.613	.167	12,020
Keenable	.879	**.721**	**.800**	**.510**	12,020
Linkup	.878	.357	.575	.130	7,898
Keenable realtime	.877	.709	.796	.509	12,020
Perplexity	.865	.585	.668	.309	11,636
Jina	.862	.629	.470	.030	11,716
Perplexity Fast	.860	.574	.666	.305	11,636
Claude search	.860	.427	.620	.182	10,400
Context	.856	.628	.516	.111	12,020
Exa Lite	.848	.371	.305	.219	420
Firecrawl	.836	.636	.472	.028	12,020
Google (Serper)	.834	.601	.457	.030	12,020
Exa	.805	.531	.462	.345	12,020
Tavily	.795	.279	.144	.033	12,020
TinyFish	.769	.563	.440	.018	3,472
Exa Instant	.713	.409	.374	.273	12,020
Octen	.660	.063	.147	.157	12,020
DeepSeek	.608	.183	.184	.109	4,820
Ceramic	.011	.000	.000	.000	12,007

Table 10: Most configurations find a paper from its title (23 of 26 at .71–.98), but body-keyword recall runs from .00 to .72. Scholar recall@10 per query bucket over the 31 runs of Figure 1a (29 September to 28 October 2026; 3,005 query-runs per bucket for the ceiling), query-weighted; failures count as in Table 1. Query-runs sum the four buckets and are fewer where a configuration missed runs or lost queries. Bold: best configuration per bucket (`bucket_table.py`).

.78 and .79 (.75–.82). Raters from three vendors therefore agree with one another more than any agrees with the GPT-6 judge, and all three rate higher than the judge (mean rating 1.88 for Gemini, 1.87 for Grok, 1.97 for the Claude consensus, 1.60 for the judge). Because the Claude raters saw the page while writing the snippet grade, a staged check (`staged_page.py`, `audit/raters/staged/`) re-graded 400 items with a fetched, unbroken page (200 news, 200 rare) in fresh Opus sessions that saw only the date, query, URL, title and snippet. That snippet-only grade differs from rater A's page grade on 25% of items (21–29%), but page grade minus snippet-only grade is only $+.05$ ($-.01$ to $+.11$), so the page adds noise to a quarter of grades but no systematic shift at this sample size. The snippet-only grade agrees with the judge, which also sees only snippets, at κ .82 (.77–.85). At the engine level the judge and the Claude consensus rank the 21 engines at Spearman .89 on news (.74–.94) and .71 on rare (.50–.94), with 37 to 76 items per engine, so the rare figure is coarse (`rater_engine_ci.py`). The absence of a net page effect is pooled over engines; the sample does not test engine-specific page bias.

List-level checks. Because six results per query cannot give a list metric, Gemini 4 Argon and Grok 4.7 graded 100 complete pools (50 news and 50 rare queries from the Table 1 runs, every top-5 list of all 21 engines, 5,698 unique results, snippet only), one item per call under the written guideline with its page rules removed (`list_sample_big.py`, `api_grader.py`, `list_analysis.py`, `list_pair_diffs.py`, `audit/lists100/`). Recomputing every engine's pooled nDCG@5 from each rater's grades ranks the 21 engines against the judge at Spearman .94 (query-resampled 95% interval .89–.97; τ_b .85) for Gemini and .94 (.87–.96; τ_b .83) for Grok, with mean absolute share differences of .07 and .06. The same five engines lead under the judge and both raters (Keenable, Exa, Perplexity, DeepSeek, Brave (LLM ctx), in varying order) and the same three trail (TinyFish, Tavily or Kagi, Ceramic). The leading paired differences keep their sign under the judge, Gemini and Grok (Keenable–Parallel $+.204$, $+.168$ and $+.137$; Keenable–Ceramic $+.53$, $+.60$ and $+.56$), but Keenable–Exa is $+.035$ ($-.024$ to $+.091$) by the judge, $+.037$ ($-.016$ to $+.089$) by Gemini and $.000$ ($-.053$ to $+.053$) by Grok, so 100 pools do not separate the top two under the judge or either rater.

Quoted rare queries. Over the Table 1 window (`rare_quoted.py`) 504 of the 1,200 rare query-runs carry a double-quoted span, and 30 are wrapped as `query: "..."` by the source agent, so they are scored on a phrase no page contains. Dropping the wrapped queries changes no rare share by more than .002. The quoted and unquoted query-runs rank engines differently. On the 672 scored unquoted query-runs (696 issued; 18 dropped for an empty ceiling and 6 for a failed judgement), Exa, Perplexity and Brave (LLM ctx) lead and Keenable is fourth (τ_b .90 against the full rare order). On the quoted query-runs, Keenable, ChatGPT search and Brave (LLM ctx) lead, and Parallel, Perplexity, Exa and Linkup lose .14 to .18 of their rare bench share (τ_b .62). The rare col-

	News	Rare	All
results	600	600	1,200
page missing for a rater	43	85	128
κ, rater A vs. B (page)	.94	.94	.94
κ, judge vs. consensus (page)	.72	.80	.77
95% CI	.67–.77	.75–.84	.74–.81
within one level, judge vs. consensus	.86	.91	.88
mean rating, judge / consensus	1.83 / 2.32	1.38 / 1.61	1.60 / 1.97
grade revised in-prompt after the page, A / B	.10 / .09	.05 / .05	.08 / .07
staged snippet-only grade $\neq$ page grade (A), 400 items	.28	.22	.25
κ, judge vs. staged snippet-only	.78	.85	.82
Spearman ρ on 21 engine means	.89	.71	.84
κ, Gemini 4 vs. consensus / judge	.91 / .74	.92 / .81	.92 / .78
κ, Grok 4.7 vs. consensus / judge	.90 / .75	.88 / .81	.90 / .79
Spearman ρ on 21 engines, Gemini / Grok vs. judge	.83 / .81	.73 / .73	.83 / .85

Table 11: Blinded raters from three vendors (two Claude Opus 5.5 sessions, Gemini 4, Grok 4.7) agree with each other more than any agrees with the production judge. A staged snippet-only pass shows that the page changes a quarter of grades with no net shift; in-prompt revisions understate that change. Quadratic-weighted κ; intervals from 1,000 query-level bootstrap resamples; consensus is the rounded mean of the two page grades.

	News	Rare
results	803	566
exact	.554	.730
within one level	.918	.995
κ, quadratic weights	.790	.898
κ, unweighted	.440	.584
Spearman ρ on ratings	.806	.930
κ, binary at ≥ 2 / ≥ 3	.64 / .70	.69 / .83
engines	23	23
Spearman ρ on shares	.903	.874
Kendall τ_b on shares	.746	.728
mean $\lvert\Delta$ share$\rvert$	.064	.061

Table 12: The second judge (Claude Opus 5.5) agrees with the production judge (GPT-6) within one level on 92% of news and 99.5% of rare results and ranks engines at Spearman .90 on news and .87 on rare. Both judges rate the same results under the same prompts, eight queries per bench.

umn thus rewards literal compliance with quoted phrases as much as retrieval. The literal gate seldom punishes a page that does contain the phrase: of the 212 quoted-span results in the rater sample with page text, 74 show the span in the title or snippet and only 8 more show it on the page alone (`quoted_on_page.py`).

Second judge. We rendered the production prompts for every unique result of eight news queries, one from each of eight runs between 30 September and 24 October 2026 (803 results; 23 of the 25 configurations active then had complete lists for at least six queries), and of eight rare queries from the eight daily runs of 22–29 October (566 unique results, 23 engines with complete lists for at least four queries with a nonzero ceiling under each judge). The second judge, Claude Opus 5.5, rated them under the same instructions in six independent sessions, without access to the production ratings. The artifacts do not store the published date the API returned, so the second judge saw title, URL and snippet only. To separate that input difference from the model difference we re-ran the production judge (same model, prompt and temperature) on the same rendered prompts without dates. On news the dateless rerun agrees with its own dated ratings exactly on 61% of results and within one level on 91% (quadratic κ .80, query-resampled 95% interval .67–.88), and its mean rating falls from 2.43 to 2.19: the published date is a large input for news. Without dates the rerun agrees with the second judge at κ .76 (.68–.82), against .79 with dates, so the model difference is about as large as the date effect. On rare, where dates seldom matter, the rerun matches the dated ratings at κ .97 and the second judge at .90 (.81–.95; `rejudge.py`, `rejudge_analysis.py`). Table 12 gives the

agreement of the second judge with the production judge (GPT-6; `aggregate_judge.py`), with engine shares recomputed for each judge against a ceiling pooled from that judge's ratings (query-resampled Spearman intervals .76–.97 on news and .47–.96 on rare; `judge_cluster_ci.py`). The second judge is stricter: its mean news rating is 2.22 against 2.43, and 68% of disagreements go that way. Its two largest news drops are Parallel Turbo and Claude search ($-.20$ each), engines whose snippets often paraphrase the event without stating it. Claude search shares a vendor with the second judge, and ChatGPT search with the judge; in the Table 1 news column, which the judge scores, the two engines rank 17th (with our credit lapse scored zero) and 12th of 21. Eight queries per bench are few, so these engine-level figures bound judge disagreement rather than estimate it precisely; as with human assessors (Voorhees, 2000), disagreement on items leaves the engine ranking largely intact. Both judges read the same precomputed query profile, so a profile that misreads a query would not show up as disagreement. An annotator (Claude Opus 5.5) therefore wrote its own profile for the 16 sampled queries before seeing the production ones: 13 production profiles match in intent, two are narrower and one is broader, and none misreads the query or states the answer.

Equal snippet budget. On the 231 Table 1 news runs with per-query artifacts, the per-engine median snippet length ranges from 117 characters (Bing) to the 2,000-character cap (Jina), and its rank correlation with news share is .34 (`snippet_length.py`, `audit/snippet_length.json`). The six engines with medians below 200 characters rank 11th to 19th of 21 on news; of the three longest, Jina and Ceramic rank 20th and 21st but DeepSeek ranks third. To bound the effect of snippet length, the second judge rated the same 803 results again with every snippet cut to 200 characters. Against its own full-snippet ratings (`aggregate_trunc.py`), exact agreement is .60, agreement within one level .94 and the quadratic κ .80. Over the 23 configurations of Table 12, Kendall τ_b between the two engine orders is .73 (query-resampled interval .40–.84; `trunc_ci.py`) and Spearman ρ .88 (.52–.95), with a mean absolute share difference of .084 (.06–.14). The cap compresses the field: the intervals of 10 of the 23 configurations exclude zero, among them the losses of Exa ($-.22$, interval $-.41$ to $-.08$), Keenable ($-.14$, $-.23$ to $-.05$) and Brave (LLM ctx) ($-.12$, $-.21$ to $-.04$) and the gains of Bing ($+.14$, $+.10$ to $+.19$) and Google ($+.09$, $+.04$ to $+.15$). The interval of the leader's margin over the second engine includes zero both with full snippets and at 200 characters, and the leader and the bottom two do not change. Eight queries thus establish the direction of the length effect but not its size per engine.

H Recall by query operator

The finance and legal generators assign operator syntax by cycling through a fixed list, so in production the variants of one bucket are not paired on the same item. To test operator effects we ran a paired set on 30 October 2026. As in the finance and legal columns of every table, a `site:` hit counts on any host, so a loss measures lost recall, not disobedience.

Paired variants (30 October 2026). To pair the variants on the same item we took 80 filing-document, 80 caselaw and 80 CFR targets from the window's finance and legal query sets, stripped every operator and quotation mark to make a plain query, and wrote the quoted, `site:` and date variants of each (`paired_build.py`; the date variant brackets the filing or decision date by 45 days). We ran the variants of each target adjacently through 18 configurations in one run per bench on 30 October 2026 (`paired_analysis.py`, `audit/paired/`; no Kagi, You.com or DeepSeek). Google and Jina returned redirect URLs (Appendix A) on 38–43% and 34–38% of their results on the two benches, and Context on 15% of its finance results, so their rows in Table 13 measure that change too. Keenable ran without an API key, so the public rate limit refused 91 of its 320 finance requests (229 answered). On the 80 filing targets the quoted span raises recall@5 by .10 on average (target-resampled 95% intervals lie above zero for 8 of 18 engines and below it for none), `site:sec.gov` by .35 (16 of 18 gain), and the 90-day date window changes it by $-.02$. On the 80 caselaw captions, `site:courtlistener.com` lowers recall@5 by .36 on average and for 15 of 18 engines beyond their intervals, from ChatGPT search ($-.88$) and TinyFish ($-.68$) to Keenable and Brave (LLM ctx) ($-.03$ each) and Ceramic (0); the quoted caption changes recall by $-.05$ and the

	SEC filings (80 targets)				Caselaw captions (80 targets)				CFR sections (80 targets)		
Engine	plain	quoted	`site:`	dates	plain	quoted	`site:`	dates	plain	quoted	`site:`
Perplexity	.66	$+.06^{\dagger}$	$+.07^{\dagger}$	$+.03^{\dagger}$	.97	$-.03^{\dagger}$	$-.07$	$0^{\dagger}$	.64	$+.14$	$+.11^{\dagger}$
Claude search	.54	$+.01^{\dagger}$	$+.20$	$+.09^{\dagger}$	.97	$-.10^{\dagger}$	$-.30$	$0^{\dagger}$	–	–	–
Parallel	.53	$+.01^{\dagger}$	$+.29$	$+.06^{\dagger}$	.96	$0^{\dagger}$	$-.14$	$+.01^{\dagger}$	.70	$+.09$	$-.01^{\dagger}$
Keenable	.52	$+.12$	$+.28$	$-.15$	.99	$-.09$	$-.03^{\dagger}$	$-.03^{\dagger}$	.64	$+.09$	$+.25$
Exa	.51	$+.16$	$+.28$	$+.14$	.96	$-.03^{\dagger}$	$-.55$	$0^{\dagger}$	.60	$+.14$	$+.21$
ChatGPT search	.45	$+.05^{\dagger}$	$+.25$	$+.05^{\dagger}$	.95	$-.25$	$-.88$	$-.03^{\dagger}$	.56	$+.15$	$+.16$
Brave (LLM ctx)	.29	$+.11$	$+.38$	$-.15$	.96	$0^{\dagger}$	$-.03^{\dagger}$	$-.23$	.53	$+.05$	$+.09^{\dagger}$
Bing (SearchAPI)	.29	$+.01^{\dagger}$	$+.39$	$+.01^{\dagger}$	.91	$-.06^{\dagger}$	$-.62$	$-.06^{\dagger}$	.51	$+.11$	$+.16$
Linkup	.26	$-.01^{\dagger}$	$+.54$	$-.16$	.93	$-.01^{\dagger}$	$-.25$	$-.33$	.45	$0^{\dagger}$	$+.14$
Brave	.23	$+.10$	$+.42$	$-.06^{\dagger}$	.97	$-.16$	$-.06$	$-.33$	.55	$0^{\dagger}$	$+.06^{\dagger}$
Firecrawl	.15	$+.30$	$+.62$	$+.04^{\dagger}$	.89	$-.03^{\dagger}$	$-.65$	$-.01^{\dagger}$	.31	$+.39$	$+.56$
Jina	.15	$+.09^{\dagger}$	$+.49$	$-.07^{\dagger}$	.61	$+.01^{\dagger}$	$-.28$	$-.16$	.25	$+.44$	$+.60$
TinyFish	.12	$+.19$	$+.47$	$-.04^{\dagger}$	.88	$-.14^{\dagger}$	$-.68$	$-.18^{\dagger}$	.21	$+.47$	$+.61$
Tavily	.12	$+.04^{\dagger}$	$+.39$	$-.07^{\dagger}$	.84	$+.04^{\dagger}$	$-.35$	$-.25$	.55	$0^{\dagger}$	$+.07^{\dagger}$
Context	.11	$+.23$	$+.44$	$+.03^{\dagger}$	.82	$+.04^{\dagger}$	$-.61$	$-.03^{\dagger}$	.30	$+.38$	$+.55$
Google (Serper)	.07	$+.24$	$+.56$	$-.03^{\dagger}$	.86	$-.10^{\dagger}$	$-.64$	$-.35$	.26	$+.47$	$+.56$
Octen	.07	$0^{\dagger}$	$+.24$	$0^{\dagger}$	.69	$+.04^{\dagger}$	$-.41$	$-.06^{\dagger}$	.56	$+.07^{\dagger}$	$+.09^{\dagger}$
Ceramic	.00	$0^{\dagger}$	$0^{\dagger}$	$0^{\dagger}$	.19	$-.01^{\dagger}$	$0^{\dagger}$	$0^{\dagger}$	.14	$0^{\dagger}$	$0^{\dagger}$

Table 13: A `site:` restriction helps on filings and CFR sections and hurts on caselaw captions; the quoted span helps on filings and CFR and costs a little on captions. This run cannot tell whether the caselaw loss comes from index coverage, ranking or the operator's implementation. Plain recall@5 and the paired change under each variant on the same targets, one run per bench on 30 October 2026; a change printed in dark grey and marked † has a target-resampled 95% interval that includes zero, the others exclude it. Google and Jina returned redirect URLs on 38–43% and 34–38% of their results, and Context on 15% of its finance results (Appendix A); Keenable's finance cells rest on 229 of 320 requests.

date window by $-.11$. On the 80 CFR sections the quoted span adds .18 (12 of 17 engines) and `site:law.cornell.edu` adds .25 (10 of 17). This run does not separate coverage, ranking and operator implementation. Failures follow the rule of Appendix A: an account-side failure leaves the target out of that engine's difference, and any other failure scores zero. In the legal run Claude search has no CFR cell and its caselaw cells rest on 29 or 30 targets (credit lapse), and Tavily's CFR cells rest on 29 targets and TinyFish's caselaw and CFR cells on 34 to 42 and 59 to 61 (rate limits). ChatGPT search's 3 failed caption requests, Bing's 1 and TinyFish's 2 failed filing requests, and Jina's HTTP 422 errors on 35 filing, 28 caption and 1 CFR request score zero. If the targets left out differ in operator effect from the rest, the estimate is biased in a way the interval does not show. Excluding the engines with redirect URLs on more than 1% of their results (Google and Jina, and Context on filings) leaves the caselaw `site:` loss at $-.35$ (13 of 16 engines beyond their intervals), the filing `site:` gain at $+.32$ and the CFR gains at $+.14$ (quoted) and $+.20$ (`site:`); restricting to the 14 engines with no failed caption request gives a caselaw `site:` loss of $-.32$.

Operator compliance. For every `site:` query of the 24 finance runs of Appendix F and the 10 legal runs of 18–28 October 2026 we measured the share of returned URLs whose host is the requested one. It is 1.00 on both benches for 16 of the 21 engines. Claude search and Google keep 1.00 on finance and .98 and .99 on legal; Bing keeps .95 and .86, Kagi .19 and .09, and Ceramic .06 and .03. The Kagi and Ceramic figures are limits of the vendor interface, not of the clients: Kagi's lens parameter (`sites_included`), its only host filter, does not restrict results, and Ceramic's API takes a bare query string, so the client strips the operators; passing them as raw text returns no results on all 20 test queries (`ceramic_rawops.py`). The stored results carry no published date, so we cannot audit date-operator compliance. Counting only `site:` hits on the requested host lowers recall on `site:` queries, pooled over every configuration's answered requests (`compliant_hits.py`), from .578 to .527 on captions, .660 to .642 on CFR and .654 to .652 on filings; only Kagi (.91 to .00 on captions) and Ceramic change, and every other engine moves by under .01.